

An introduction to p -value functions

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My promises

After this presentation, you will ...

- know what p -value functions are
- know how p -value functions are constructed
- be able to interpret p -value functions correctly
- be able to use p -value functions to improve statistical and quantitative reasoning
- be able to use p -value functions to compare study results

Research has a problem

Open access, freely available online

Essay

Why Most Published Research Findings Are False

John P. A. Ioannidis

Summary

There is increasing concern that most current published research findings are false. The probability that a research claim is true may depend on study power and bias, the number of other studies on the same question, and, importantly, the ratio of true to no relationships among the relationships probed in each scientific field. In this framework, a research finding is less likely to be true when the studies conducted in a field are smaller; when effect sizes are smaller; when there is a greater number and lesser preselection of tested relationships; where there is greater flexibility in designs, definitions, outcomes, and analytical modes; when there is greater financial and other interest and prejudice; and when more teams are involved in a scientific field in chase of statistical significance. Simulations show that for most study designs and settings, it is more likely for a research claim to be false than true. Moreover, for many current scientific fields, claimed research findings may often be simply accurate measures of the prevailing bias. In this essay, I discuss the implications of these problems for the conduct and interpretation of research.

factors that influence this problem and some corollaries thereof.

Modeling the Framework for False Positive Findings

Several methodologists have pointed out [9–11] that the high rate of nonreplication (lack of confirmation) of research discoveries is a consequence of the convenient, yet ill-founded strategy of claiming conclusive research findings solely on the basis of a single study assessed by formal statistical significance, typically for a p -value less than 0.05. Research is not most appropriately represented and summarized by p -values, but, unfortunately, there is a widespread notion that medical research articles

It can be proven that most claimed research findings are false.

should be interpreted based only on p -values. Research findings are defined here as any relationship reaching formal statistical significance, e.g., effective interventions, informative predictors, risk factors, or associations. “Negative” research is also very useful. “Negative” is actually a misnomer, and

is characteristic of the field and can vary a lot depending on whether the field targets highly likely relationships or searches for only one or a few true relationships among thousands and millions of hypotheses that may be postulated. Let us also consider, for computational simplicity, circumscribed fields where either there is only one true relationship (among many that can be hypothesized) or the power is similar to find any of the several existing true relationships. The pre-study probability of a relationship being true is $R/(R+1)$. The probability of a study finding a true relationship reflects the power $1 - \beta$ (one minus the Type II error rate). The probability of claiming a relationship when none truly exists reflects the Type I error rate, α . Assuming that c relationships are being probed in the field, the expected values of the 2×2 table are given in Table 1. After a research finding has been claimed based on achieving formal statistical significance, the post-study probability that it is true is the positive predictive value, PPV. The PPV is also the complementary probability of what Wacholder et al. have called the false positive report probability [10]. According to the 2×2 table, one gets $PPV = (1 - \beta)R/(R$

<https://doi.org/10.1371/journal.pmed.0020124>

Retire statistical significance

Valentin Amrhein, Sander Greenland, Blake McShane and more than 800 signatories call for an end to hyped claims and the dismissal of possibly crucial effects.

correspondence

Remove, rather than redefine,
statistical significance

EDITORIAL

Moving to a World Beyond “ $p < 0.05$ ”

2. Don't Say “Statistically Significant”

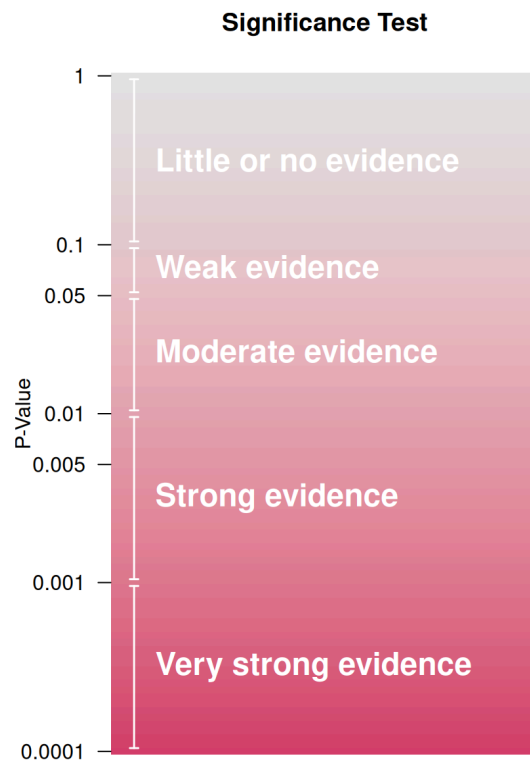
Many alternatives to classical frequentist statistics and hypothesis testing in particular have been proposed

- Bayesian statistics
- Lowering significance threshold from 0.05 to 0.005
- Focus on effect sizes and uncertainty intervals
- *My opinion: P -values are here to stay*
 - Improve interpretation of tools that are used
 - P -value functions: Graphical tool to achieve that

P-values

! Definition

P is the probability under the null hypothesis of obtaining a test statistic at least as extreme as the one obtained. “Extreme” means farther from the null value.



Informally, p -values are a continuous measure of *compatibility* between the data and a hypothesis, given a set of background information (for details, see [here](#)).

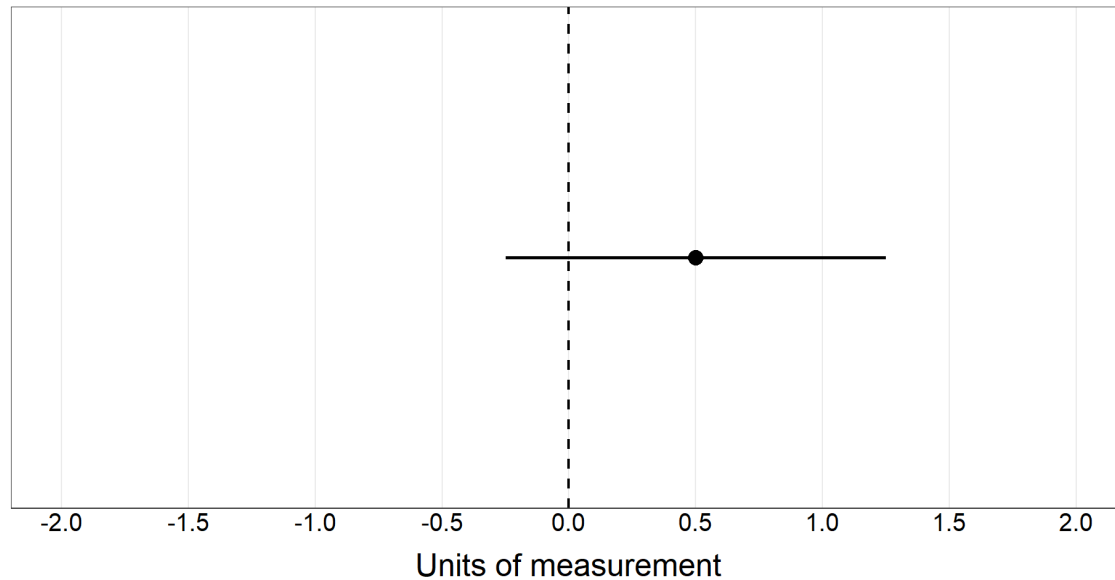
Image adapted [from here](#).

Just one p -value?

- “The null hypothesis states that there is no effect/no difference.” No!
- P -values in papers often have implicit hypotheses: $H_0 : \theta = 0$ vs. $H_1 : \theta \neq 0$.
- In other words: The p -value is for a test against the *null value* of 0.
- But: You can calculate p -values for other null values too!
- Example: $H_0 : \theta = 2.5$ vs. $H_1 : \theta \neq 2.5$.

Confidence intervals and their relation to p -values

► Code



95% Confidence interval for an arbitrary parameter

- The confidence interval (CI) displays the values that are compatible with the data and the model.
- All values within the CI have $p > 0.05$ (are not “significant”).
- Values outside the CI have $p \leq 0.05$ (are “significant”).
- **Q:** What’s the p -value for the values at the edge of the CI?
 $p = 0.05$ (for a 95%-CI)

Sitting behaviour is not associated with incident diabetes over 13 years: the Whitehall II cohort study

Emmanuel Stamatakis,^{1,2} Richard M Pulsford,³ Eric J Brunner,² Annie R Britton,² Adrian E Bauman,¹ Stuart JH Biddle,⁴ Melvyn Hillsdon³

	n/cases	Person years (×1000)	Rate (per 1000 per year)	Model 4 [†] HR 95% CI
Total sitting (hours/week)				
0 to <33	1671/145	21.11	6.87	1
≥33 to <50	1498/103	19.77	5.20	0.86 (0.66 to 1.12)
≥50	1642/154	21.58	7.14	1.19 (0.92 to 1.55)

Not significant ($p = 0.22$)

<https://doi.org/10.1136/bjsports-2016-096723>

Question

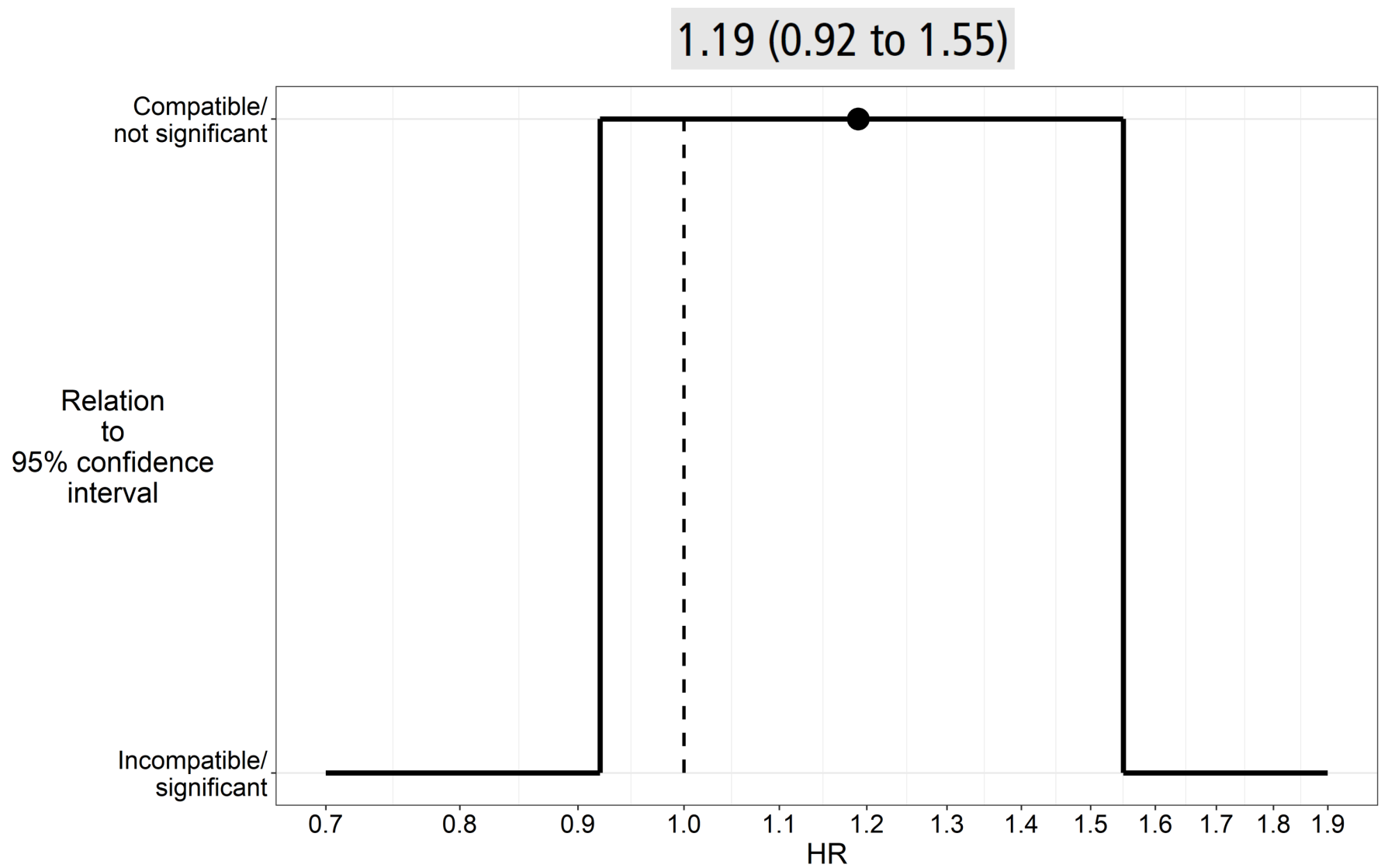
In your opinion, do the results justify the conclusion that there was *no association* between sitting time and diabetes? Why or why not?

	n/cases	Person years (×1000)	Rate (per 1000 per year)	Model 4 [†] HR 95% CI
Total sitting (hours/week)				
0 to <33	1671/145	21.11	6.87	1
≥33 to <50	1498/103	19.77	5.20	0.86 (0.66 to 1.12)
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Confidence intervals

- Confidence intervals: Range of supported/compatible values given the data.
- But:
 - Are all values inside the CI equally well supported?
 - Are all values outside the CI equally badly supported?
 - Why does the support of values decrease sharply at the edges of the CI?

95% confidence interval for Stamatakis et al. (2017)

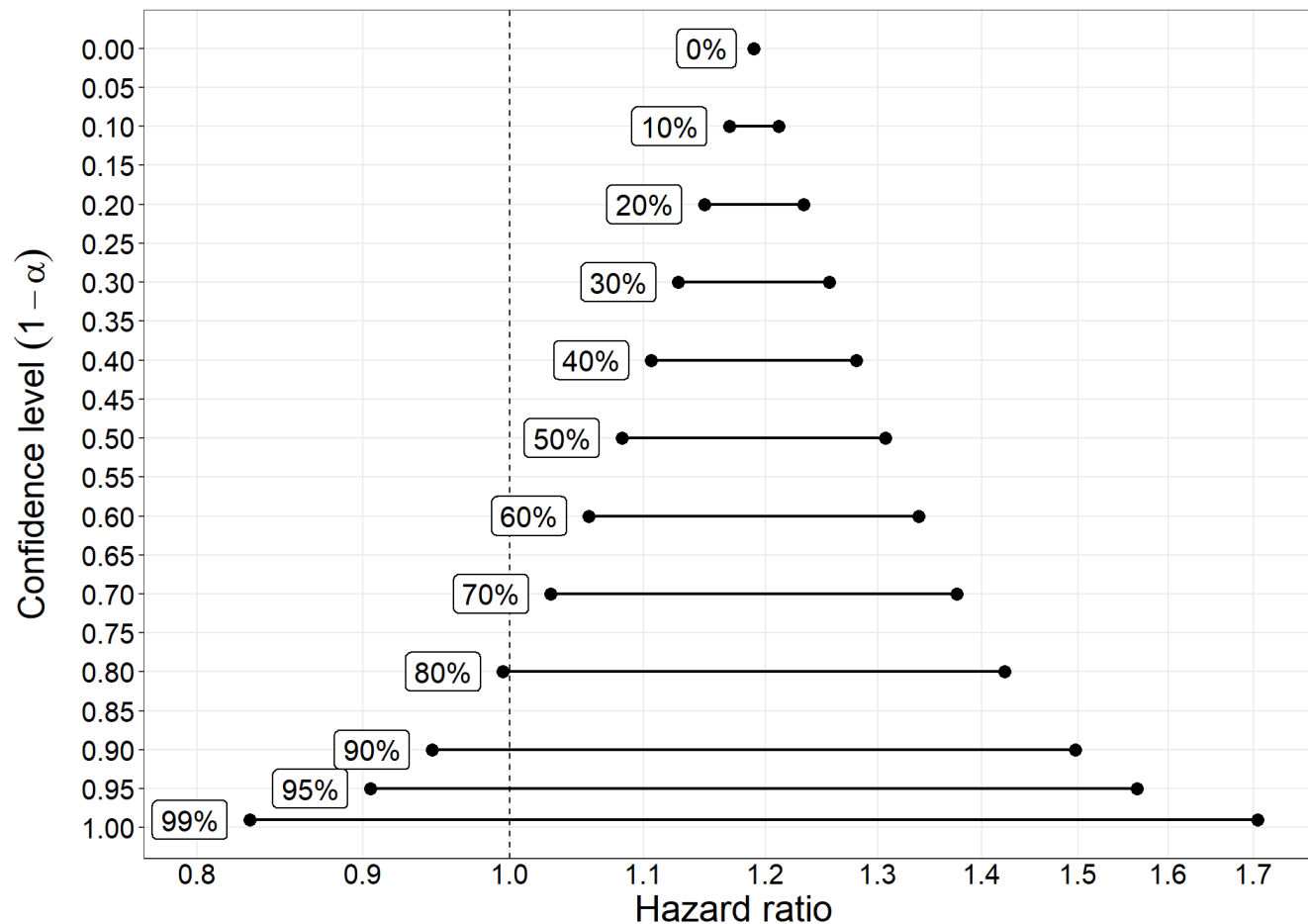


Some ideas for improvement

- Plot multiple confidence intervals together:
 - 99%
 - 95%
 - 90%
 - ...
- Calculate and plot p -values for other values than 1 (the usual null value for ratio measures):
 - For example: P -value for $H_0 : HR = 1.5$ is 0.0975.

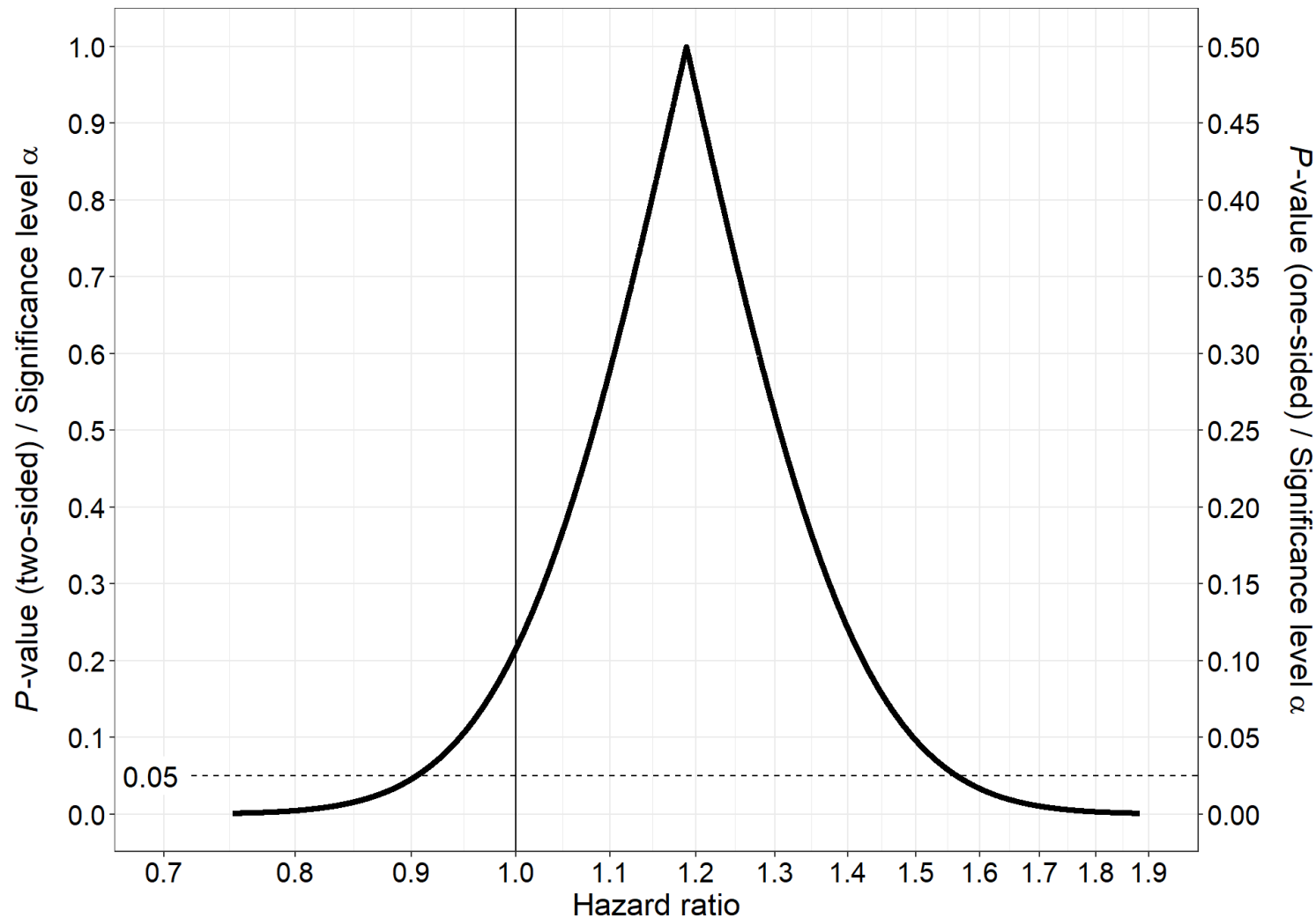
Plot multiple confidence intervals for the same estimate

► Code

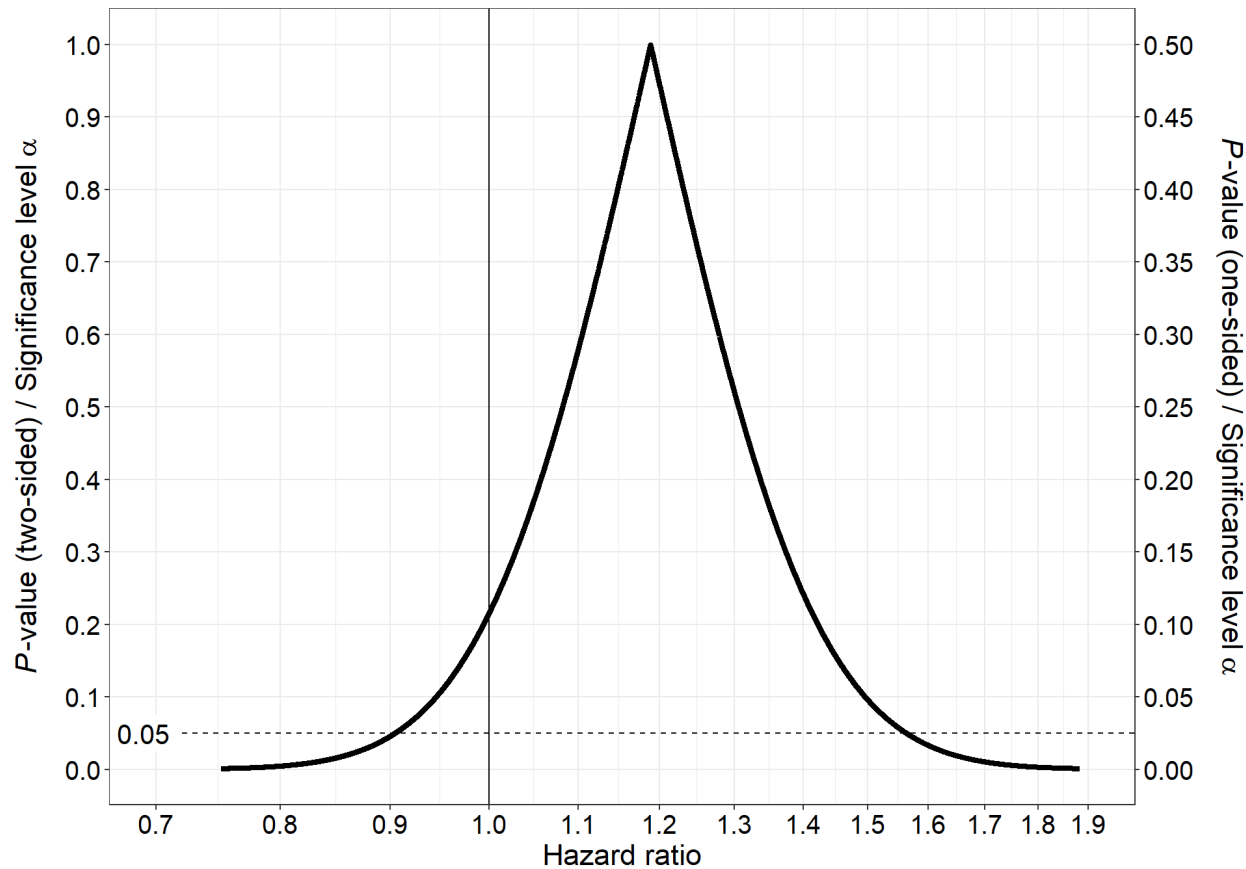


Plot *all* confidence intervals, calculate *all* p -values

► Code

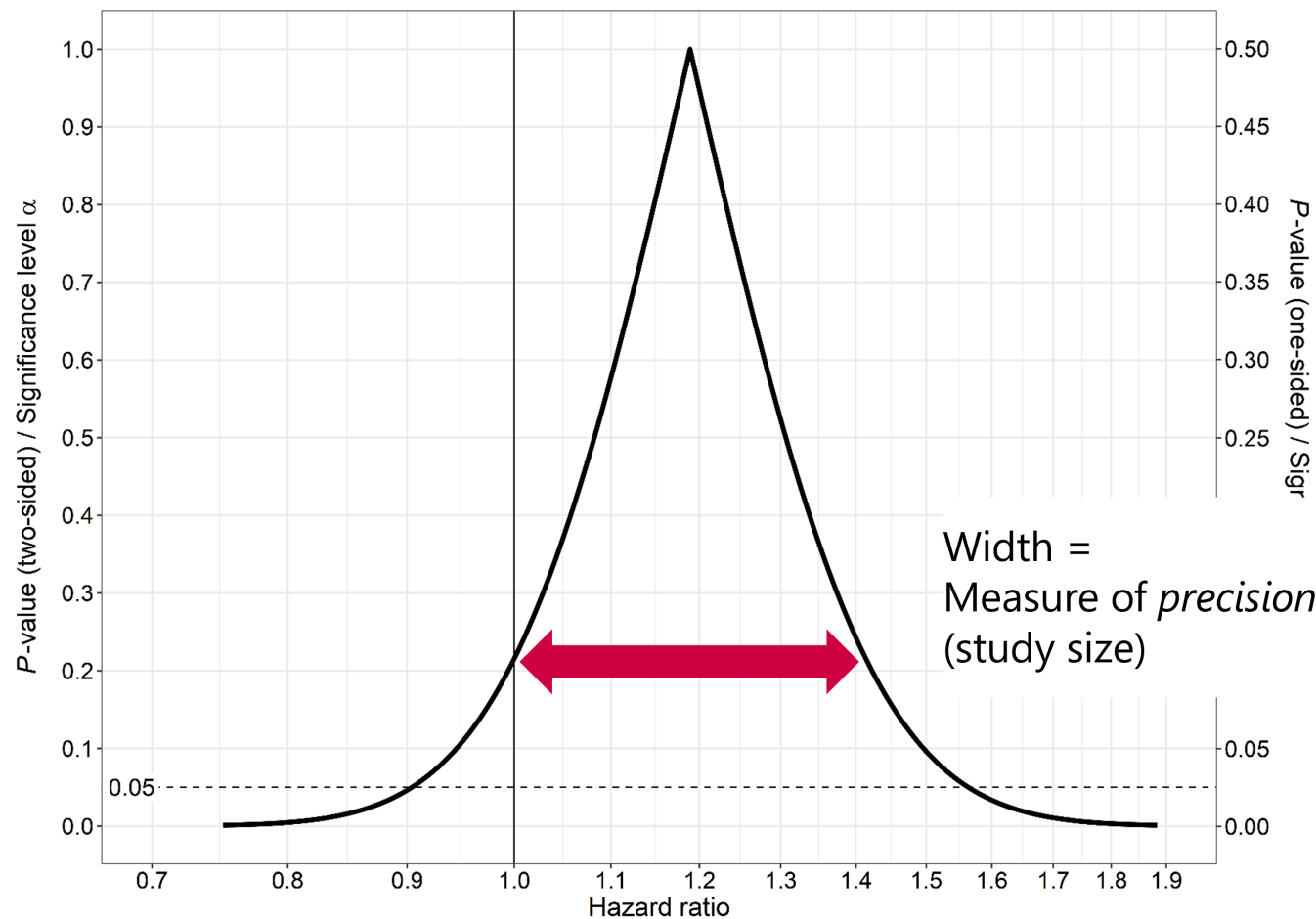


A p -value function

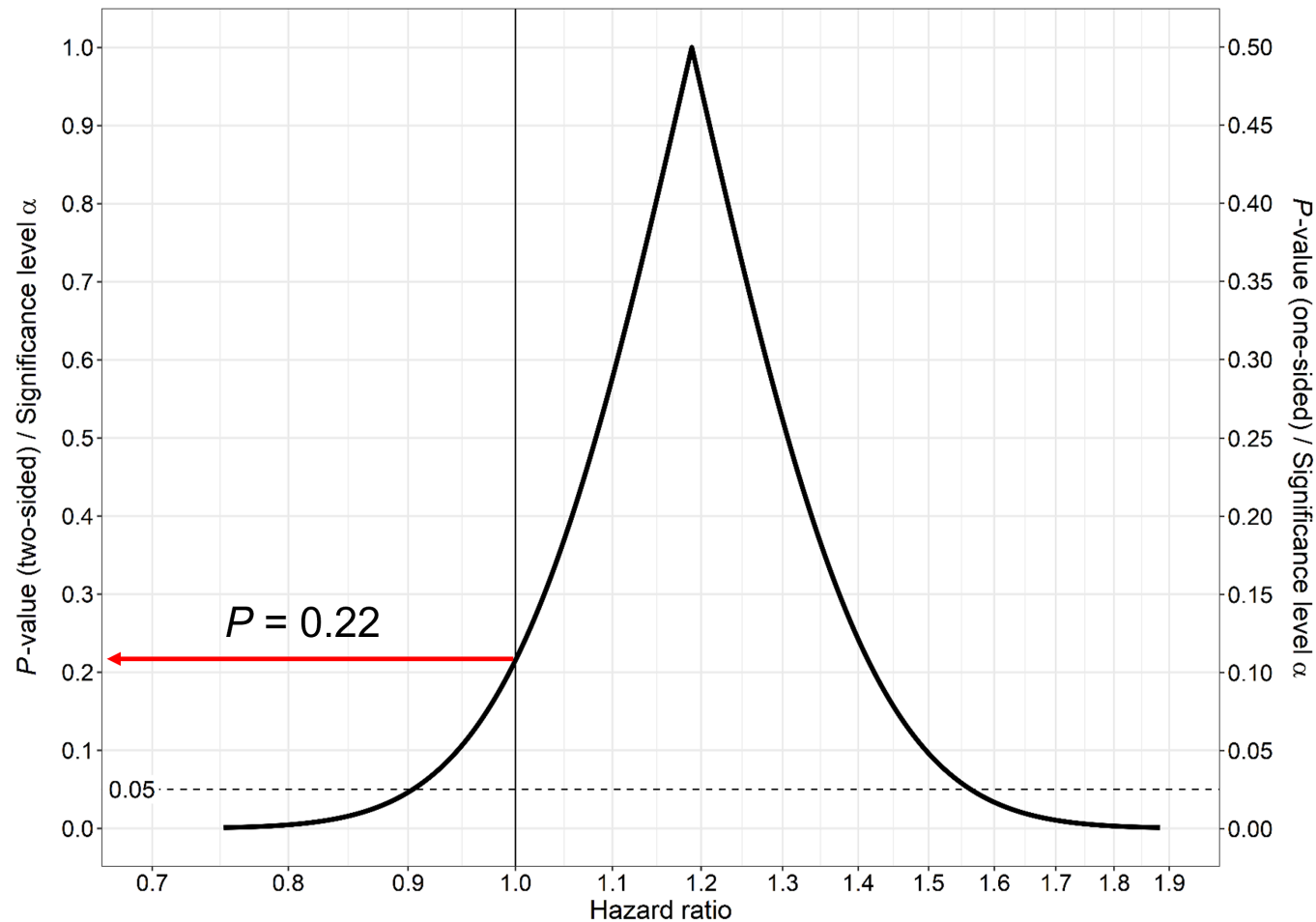


- All possible confidence intervals (all confidence levels)
- All possible p -values (two- and one-sided)

The width of the p -value function indicates the *precision* of the estimate



We can read off the p -value for any null value from the p -value function



Null misinterpretation

A null-hypothesis P value greater than 0.05 means that no effect was observed, or that absence of an effect was shown or demonstrated. No! Observing $P > 0.05$ for the null hypothesis only means that the null is one among the many hypotheses that have $P > 0.05$. Thus, unless the point estimate (observed association) equals the null value exactly, it is a mistake to conclude from $P > 0.05$ that a study found “no association” or “no evidence” of an effect. If the null P value is less than 1 some association must be present in the data, and one must look at the point estimate to determine the effect size most compatible with the data under the assumed model.

<https://doi.org/10.1007/s10654-016-0149-3>

A better summary

	n/cases	Person years (×1000)	Rate (per 1000 per year)	Model 4 [†] HR 95% CI
Total sitting (hours/week)				
0 to <33	1671/145	21.11	6.87	1
≥33 to <50	1498/103	19.77	5.20	0.86 (0.66 to 1.12)
≥50	1642/154	21.58	7.14	1.19 (0.92 to 1.55)

“Our results are most compatible at the 95% level with an effect of high versus low amounts of sitting anywhere from an 8% hazard reduction to 55% increase in the hazard of diabetes.”

Safety of ibuprofen in infants younger than six months: A retrospective cohort study

Paul Walsh^{1*}, Stephen J. Rothenberg², Heejung Bang³

Results

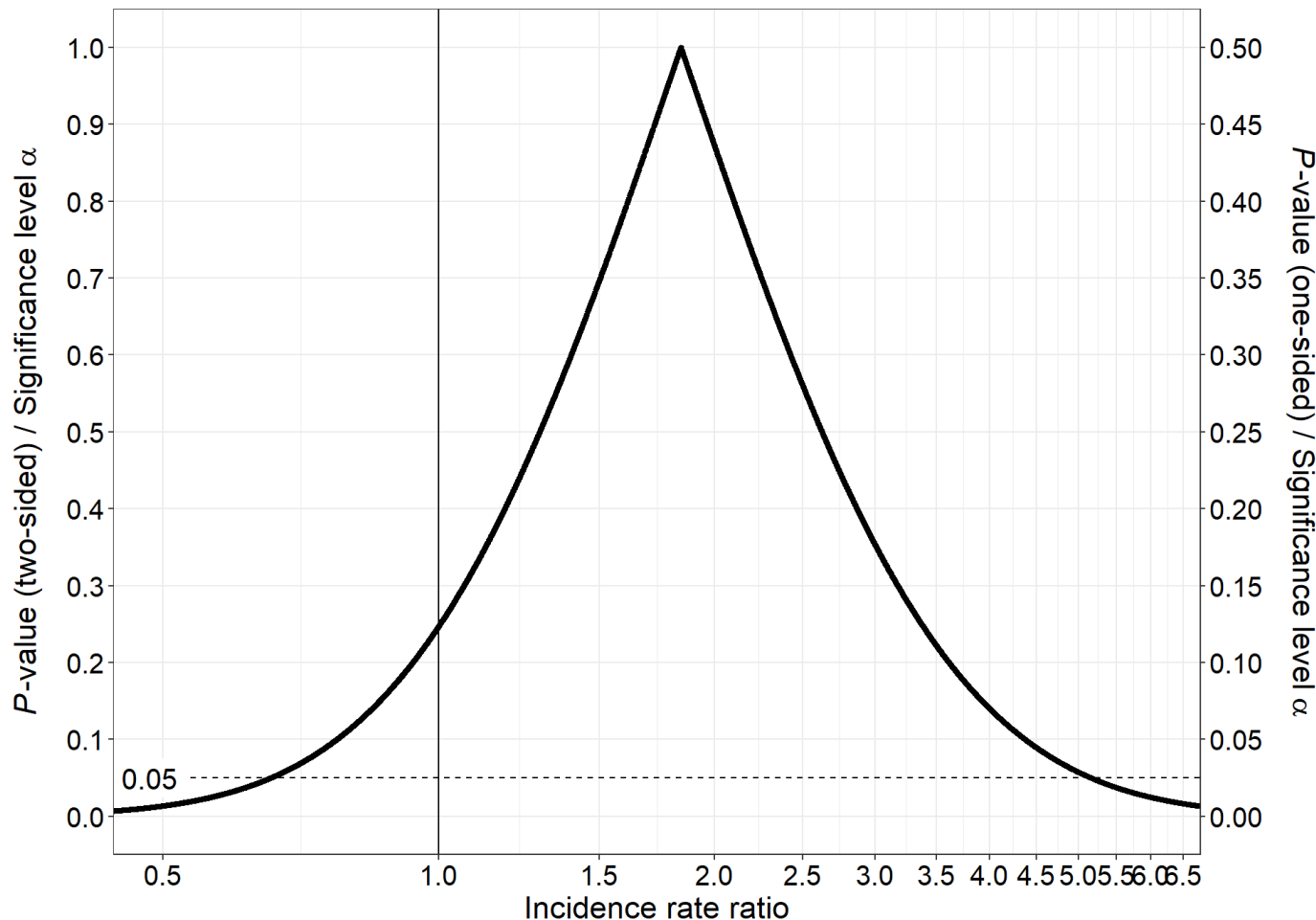
Severe GI (aRR 0.63, 95% CI 0.27–1.45) and renal AE (aRR 1.84 95% CI 0.66–5.19) were not different between the ibuprofen (+/-acetaminophen) and acetaminophen-only groups.

<https://doi.org/10.1371/journal.pone.0199493>

Implication: Ibuprofen is safe in terms of renal adverse effects (AE).

Walsh et al. (2018): Safety of Ibuprofen

► Code



How can we find the p -value of the hypothesis $H_0 : \text{IRR} = 1$ from the plot? $p \approx 0.25$

Q: How would you summarize the results?

Walsh et al. (2018): Better conclusion

- The region of high (95%) compatibility for the incidence rate ratio: 0.66 to 5.19.
- Anything from a 34% rate reduction up to a fivefold increase!

“Our study lacked sufficient information to reach useful inference about adverse renal events comparing ibuprofen to acetaminophen alone.”

Does high power plus nonsignificance support the null?

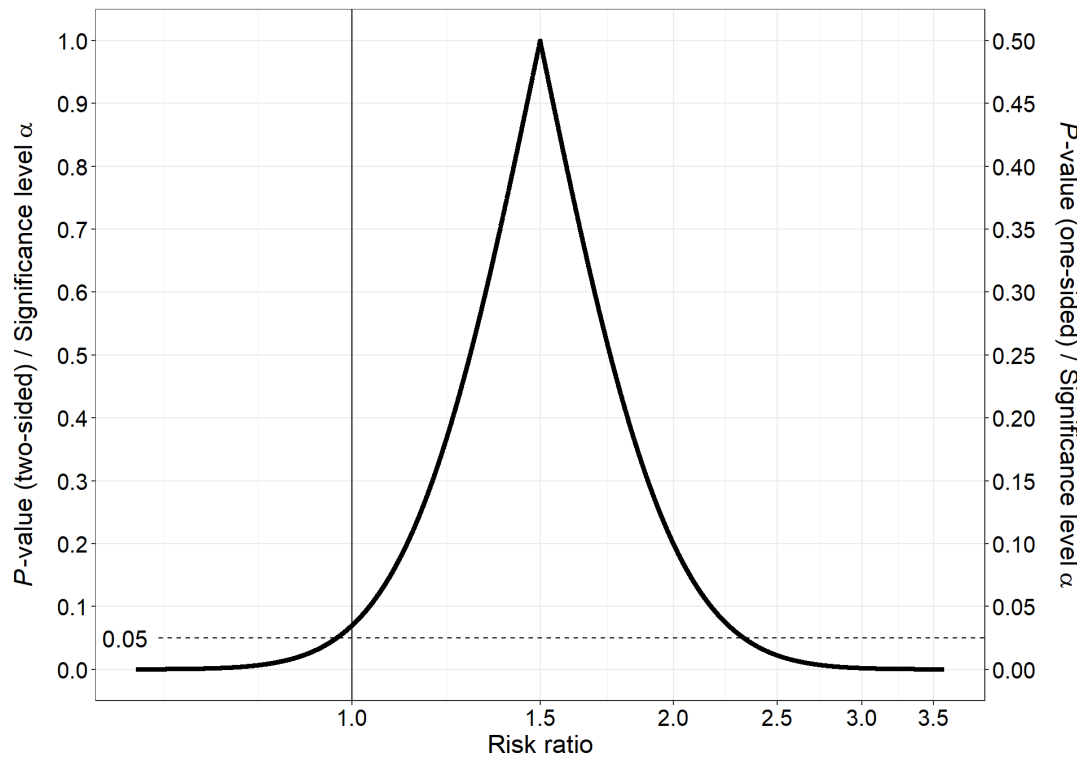
Table 1: Hypothetical randomized trial data

	New treatment	Placebo
Adverse events	48	32
Total	1000	1000

- High power for $RR = 2$ at $\alpha = 0.05$ (over 85%)
- $\widehat{RR} = 1.50$ (95%-CI: 0.97 to 2.33)
- $P = 0.07$ for $H_0 : RR = 1$
- Does this mean that the study supports the conclusion that new treatment is equal to placebo in terms of AEs? **No!**

Does high power plus nonsignificance support the null?

► Code



Based on the results, do the data favor $RR = 1$ or $RR = 2$?

- The point estimate is 1.5: Closer to 2 than to 1 in proportional terms.
- P -value for $RR = 2$ is 0.20, 3 times the p -value for $RR = 1$.
- 1 is closer to lower CI limit than 2 is to the upper limit (in proportional terms).

Summary: Despite “nonsignificance” and power approximating 90% for $RR = 2$, the results favor $RR = 2$ over $RR = 1$! ([Greenland, 2012](#))

Comparing studies using p -value functions

ORIGINAL ARTICLE

Effects of Moderate Alcohol Consumption on Cognitive Function in Women

Meir J. Stampfer, M.D., Jae Hee Kang, Sc.D., Jennifer Chen, M.P.H.,
Rebecca Cherry, M.D., and Francine Grodstein, Sc.D.

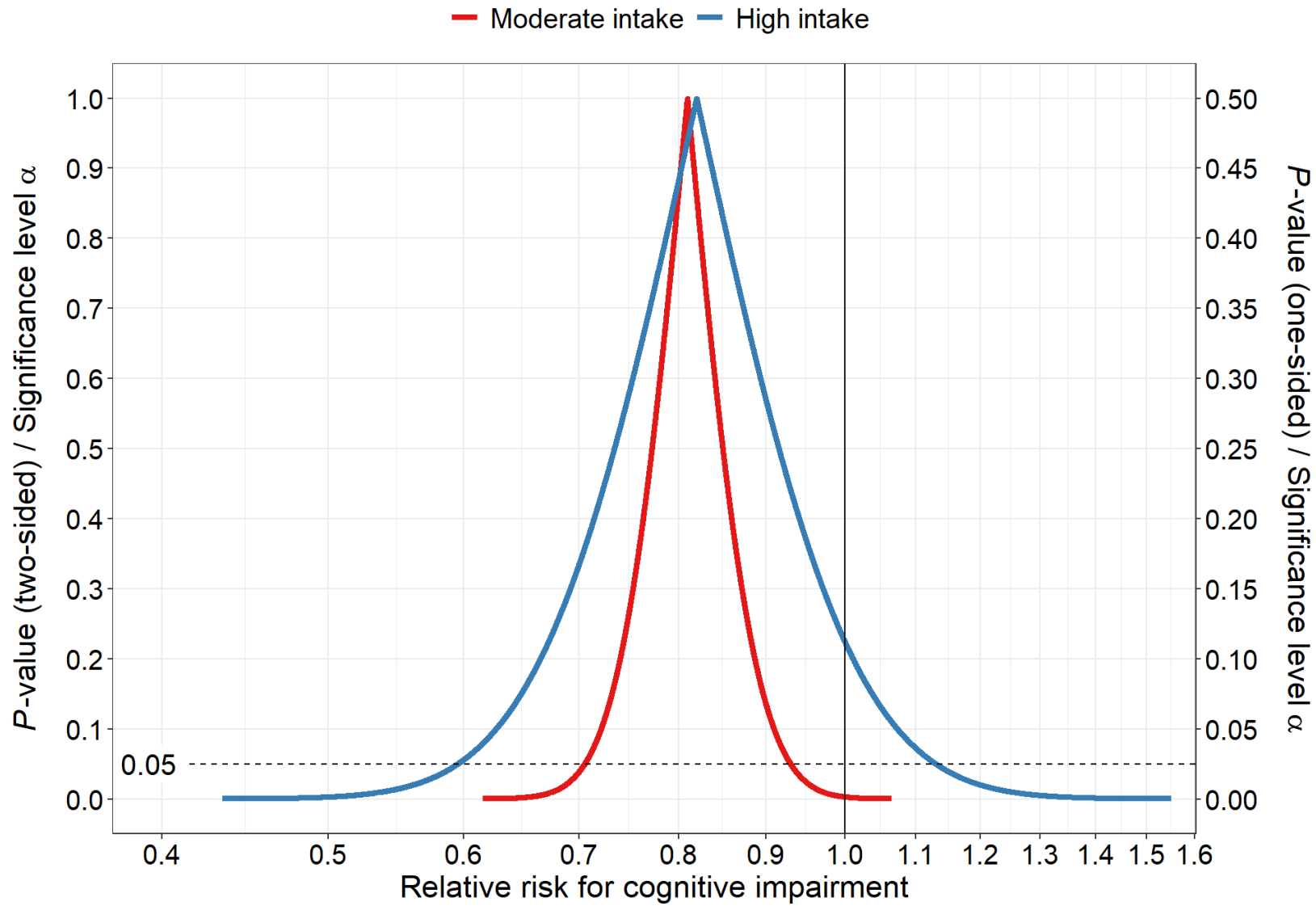
RESULTS

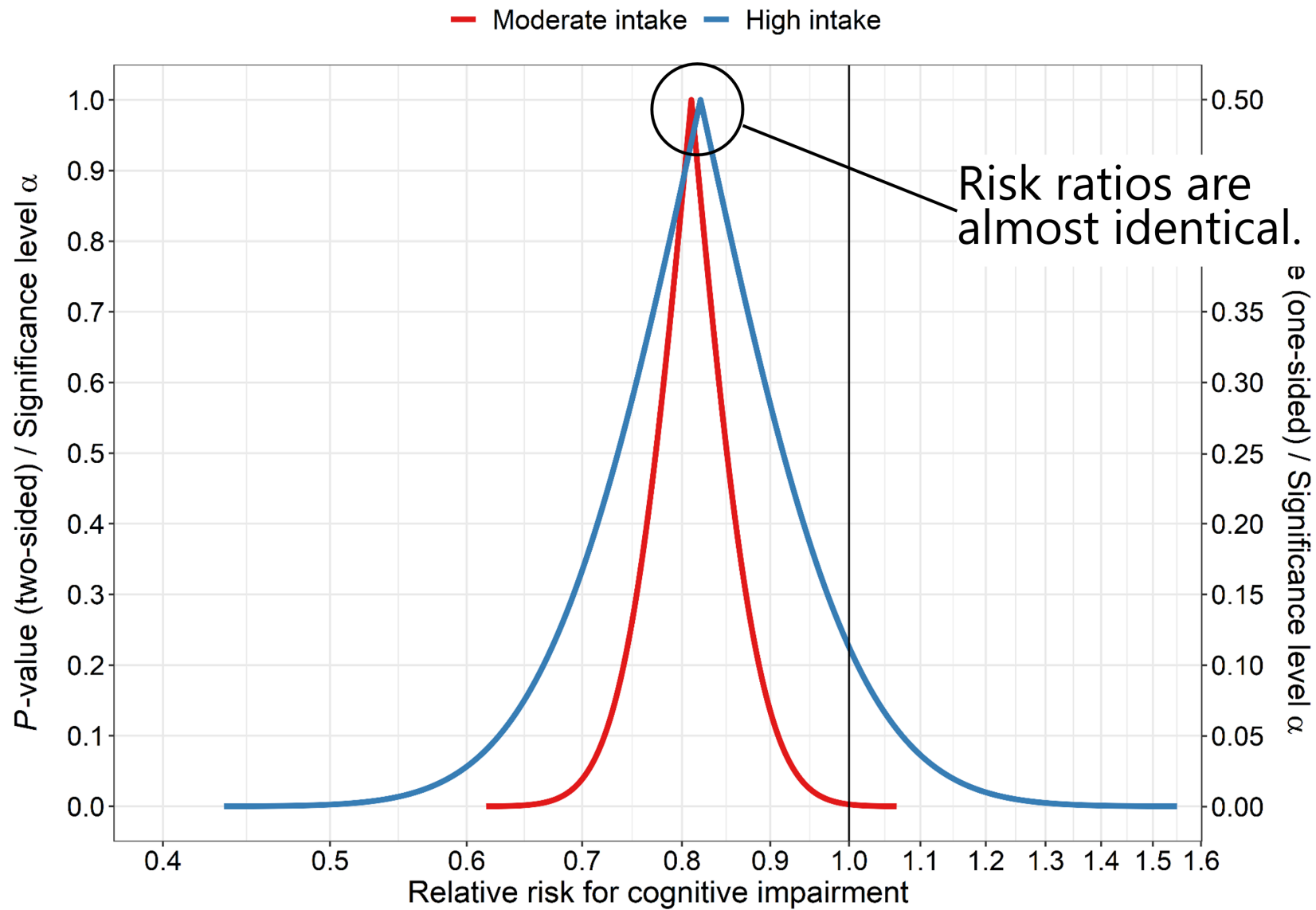
Among moderate drinkers, as compared with nondrinkers, the relative risk of impairment was 0.81 on the basis of a global cognitive score combining the results of all tests (95 percent confidence interval, 0.70 to 0.93).

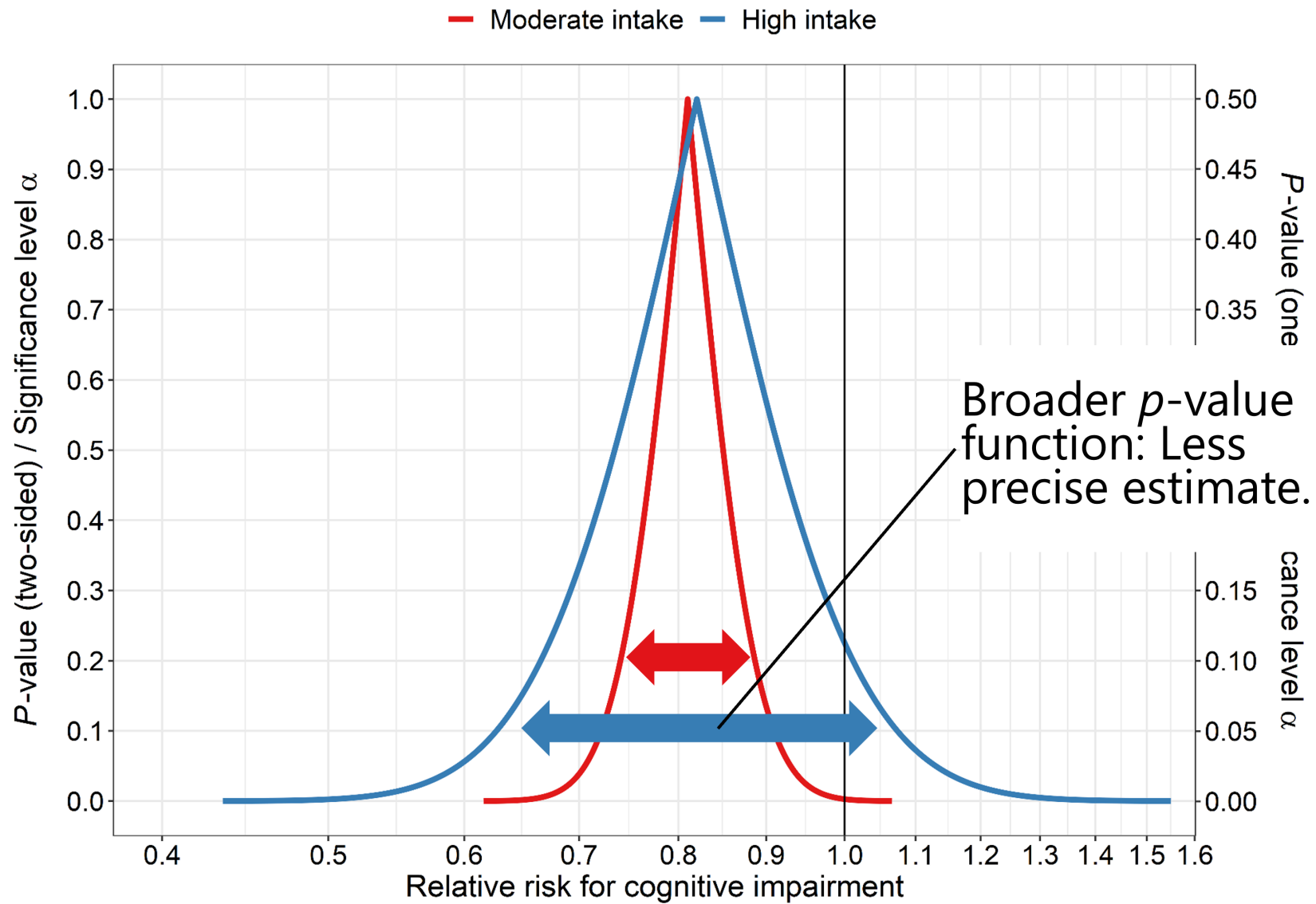
There were no significant associations between higher levels of drinking (15.0 to 30.0 g per day) and the risk of cognitive impairment or decline.

<https://doi.org/10.1056/nejmoa041152>

Implication: No association for women with high alcohol consumption or effect differs between moderate and high consumption.







Stampfer et al. (2005): A better summary

- No basis to infer that the effect size differs for moderate and heavy drinkers.
- The hypothesis that is most compatible with the data is almost the same in both groups.
- Lack of significance was driven by lower precision for heavy drinkers.

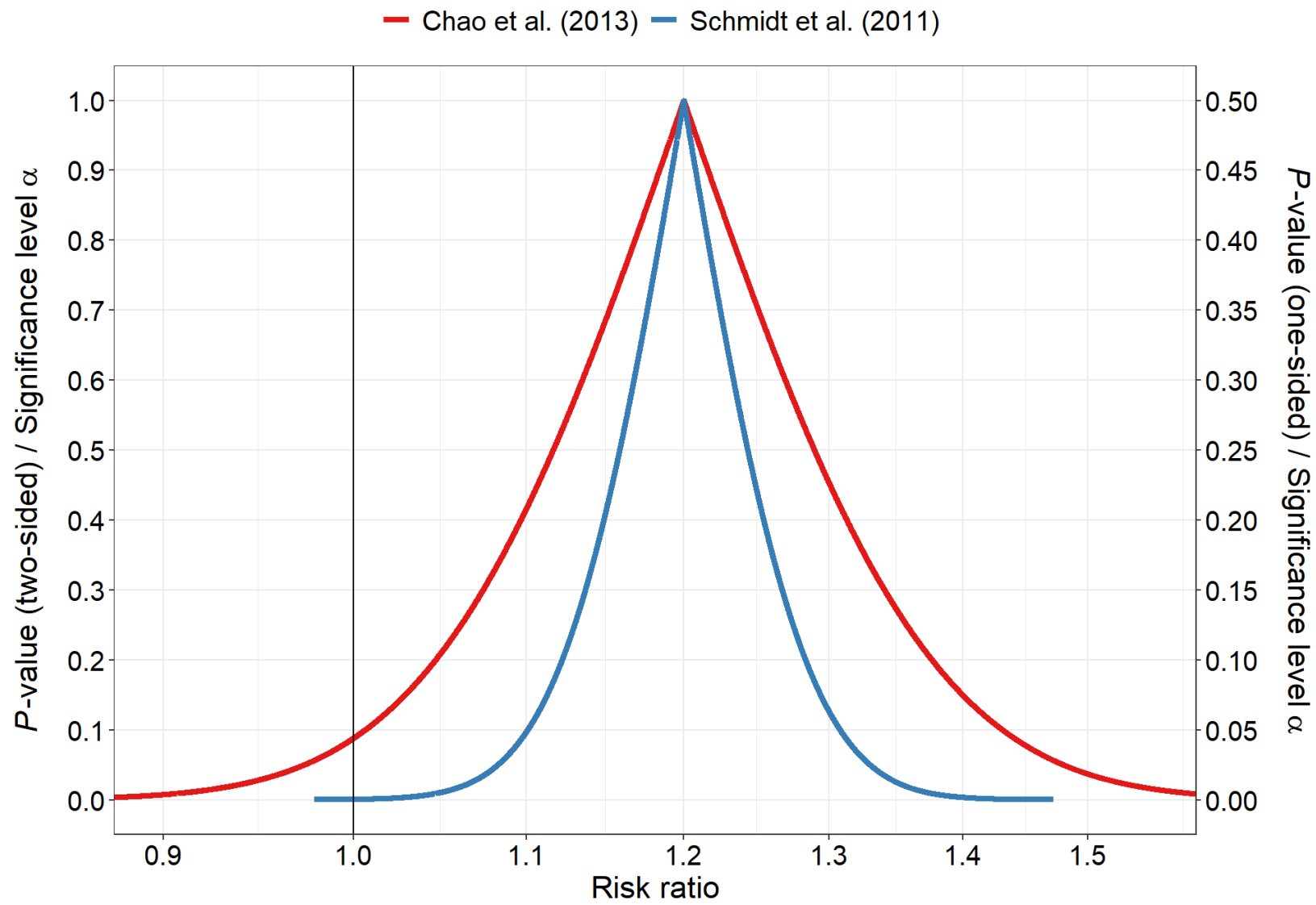
NSAIDs and atrial fibrillation risk

Previously, Schmidt et al. [2] reported that the use of non-aspirin NSAIDs was associated with an increased risk of atrial fibrillation or flutter and that the association was strongest for new users.

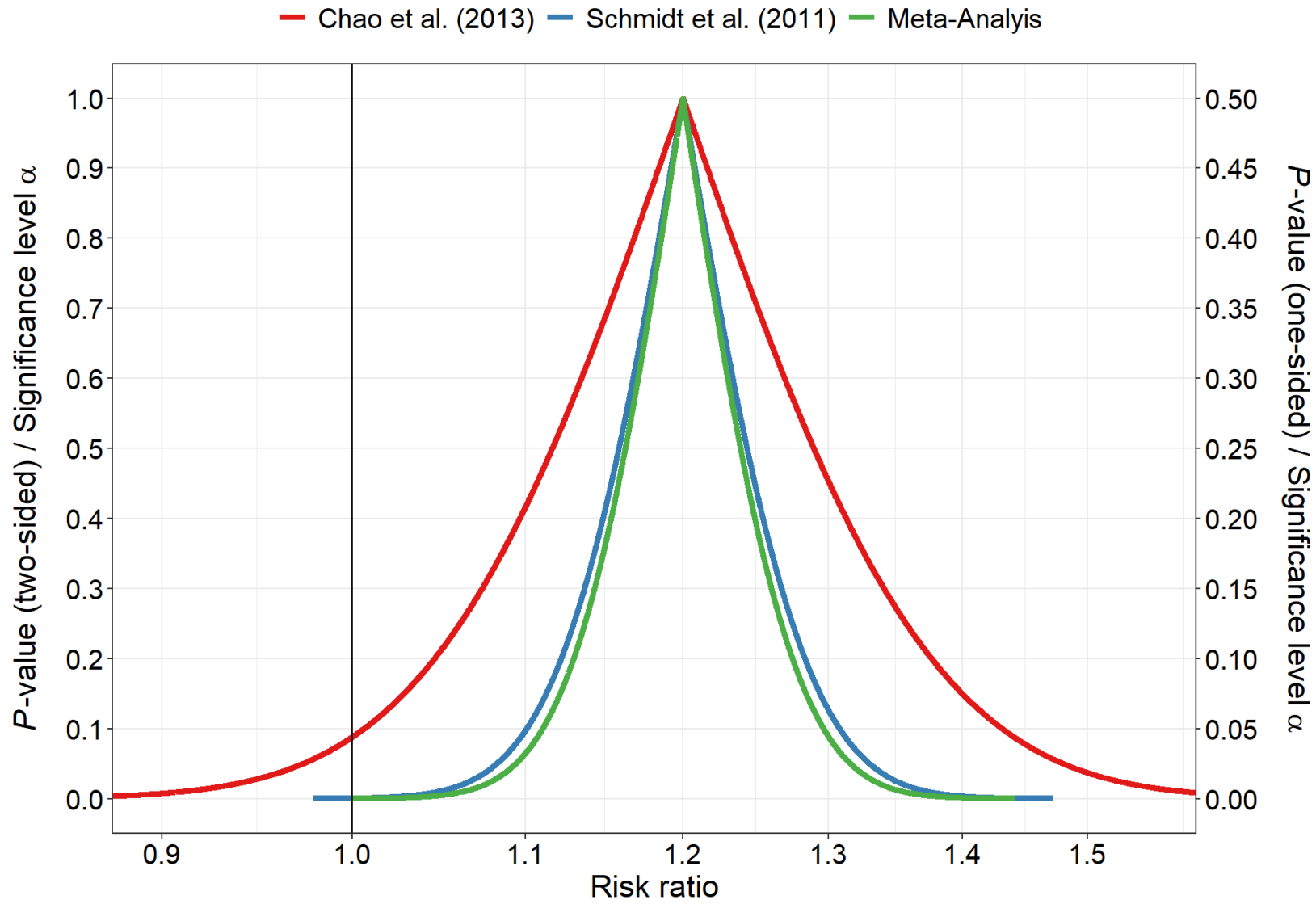
In apparent contrast to the results of Schmidt et al. [2], Chao et al. reported that “a unique finding of their study was that the use of selective COX-2 inhibitors was not associated with atrial fibrillation risk, except in patients with chronic renal or pulmonary disease” [3].

<https://doi.org/10.1016/j.ijcard.2014.09.205>

- Schmidt et al. (2011): RR = 1.20 (95%-CI: 1.09 to 1.33) (“significant”)
- Chao et al. (2013): RR = 1.20 (95%-CI: 0.97 to 1.48) (“not significant”)



► Code



Summary p -value functions

- Display all possible confidence intervals/ p -values
- Tip: Point estimate
- Width: Determined by standard error (precision)
- Standard error can be extracted from published confidence intervals or from p -value if not provided.
- Aliases: Confidence curve, compatibility curve, consonance curve, consonance function.

Literature

***P* value functions: An underused method to present research results and to promote quantitative reasoning**

Denis Infanger  | Arno Schmidt-Trucksäss

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Department of Sport, Exercise and Health,
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Null hypothesis significance testing has received a great amount of attention in recent years in the light of the reproducibility crisis of science. Recently, there have been calls to retire the dichotomization of study results into “significant” or “not significant” depending on whether the *P* value crosses some threshold or not. Ways of improving the interpretation of *P* values and confidence intervals are therefore needed. We illustrate the use of *P* value functions, which display confidence limits and *P* values for any confidence level for a parameter. *P* value functions accessibly display a wealth of information: point estimate for the parameter, one-sided and two-sided confidence limits at any level, and one-sided and two-sided *P* values for any null and non-null value and the counternull value. Presenting several recent examples from the literature, we show how *P* value functions can be applied to present evidence and to make informed statistical inferences without resorting to dichotomization. We argue that *P* value functions are more informative than commonly used summaries of study results such as single *P* values or confidence intervals. *P* value functions require minimal retraining, are easily interpreted, and show potential to fix many of the common misinterpretation of *P* values and confidence intervals. To facilitate the adoption of *P* value functions, we released an R package for creating *P* value functions for several commonly used estimates.

KEYWORDS

confidence interval, graphics, hypothesis testing, *P* value, statistical significance

Software: R package `pvaluefunctions`

`pvaluefunctions`: Creates and Plots P-Value Functions, S-Value Functions, Confidence Distributions and Confidence Densities

Contains functions to compute and plot confidence distributions, confidence densities, p-value functions and s-value (surprisal) functions for several commonly used estimates. Instead of just calculating one p-value and one confidence interval, p-value functions display p-values and confidence intervals for many levels thereby allowing to gauge the compatibility of several parameter values with the data. These methods are discussed by Infanger D, Schmidt-Trucksäss A. (2019) <[doi:10.1002/sim.8293](https://doi.org/10.1002/sim.8293)>; Poole C. (1987) <[doi:10.2105/AJPH.77.2.195](https://doi.org/10.2105/AJPH.77.2.195)>; Schweder T, Hjort NL. (2002) <[doi:10.1111/1467-9469.00285](https://doi.org/10.1111/1467-9469.00285)>; Bender R, Berg G, Zeeb H. (2005) <[doi:10.1002/bimj.200410104](https://doi.org/10.1002/bimj.200410104)>; Singh K, Xie M, Strawderman WE. (2007) <[doi:10.1214/0749217070000000102](https://doi.org/10.1214/0749217070000000102)>; Rothman KJ, Greenland S, Lash TL. (2008, ISBN:9781451190052); Amrhein V, Trafimow D, Greenland S. (2019) <[doi:10.1080/00031305.2018.1543137](https://doi.org/10.1080/00031305.2018.1543137)>; Greenland S. (2019) <[doi:10.1080/00031305.2018.1529625](https://doi.org/10.1080/00031305.2018.1529625)> and Rafi Z, Greenland S. (2020) <[doi:10.1186/s12874-020-01105-9](https://doi.org/10.1186/s12874-020-01105-9)>.

Version: 1.6.2
Depends: R (≥ 3.5.0)
Imports: [ggplot2](#) (≥ 3.3.0), [scales](#) (≥ 1.1.1), [zipfR](#) (≥ 0.6-70), [pracma](#) (≥ 2.3.3), [gsl](#) (≥ 2.1-7.1), stats
Suggests: [knitr](#), [rmarkdown](#), [devtools](#)
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Author: Denis Infanger  [aut, cre]
Maintainer: Denis Infanger <denis.infanger@unibas.ch>
License: [GPL-3](#)
URL: <https://github.com/DInfanger/pvaluefunctions>
NeedsCompilation: no
Materials: [README](#), [NEWS](#)
CRAN checks: [pvaluefunctions results](#)

Documentation:

Reference manual: [pvaluefunctions.html](#), [pvaluefunctions.pdf](#)
Vignettes: [P-value functions](#) (source, R code)

Downloads:

Package source: [pvaluefunctions 1.6.2.tar.gz](#)
Windows binaries: r-devel: [pvaluefunctions 1.6.2.zip](#), r-release: [pvaluefunctions 1.6.2.zip](#), r-oldrel: [pvaluefunctions 1.6.2.zip](#)
macOS binaries: r-release (arm64): [pvaluefunctions 1.6.2.tgz](#), r-oldrel (arm64): [pvaluefunctions 1.6.2.tgz](#), r-release (x86_64): [pvaluefunctions 1.6.2.tgz](#), r-oldrel (x86_64): [pvaluefunctions 1.6.2.tgz](#)
Old sources: [pvaluefunctions archive](#)



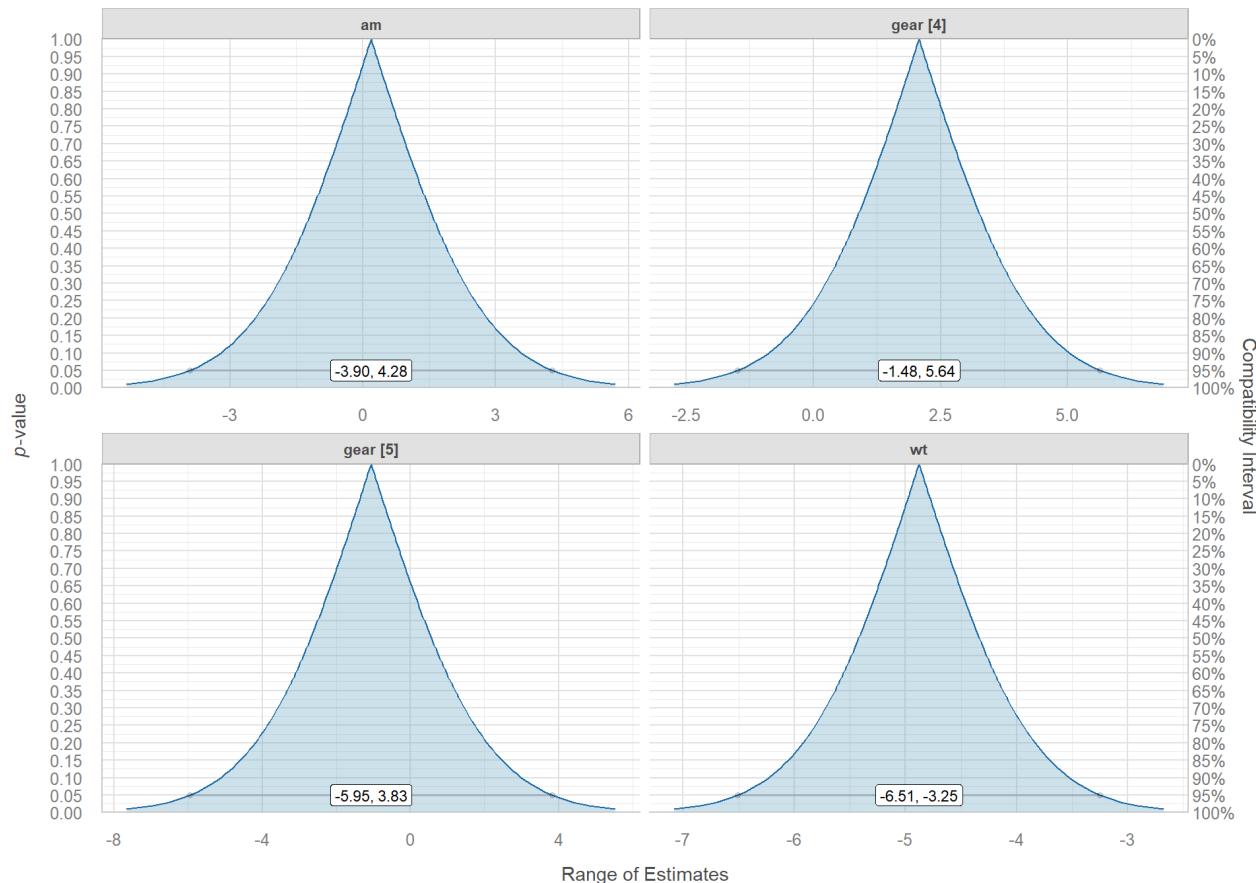
<https://cran.r-project.org/package=pvaluefunctions>

<https://github.com/DInfanger/pvaluefunctions>

Software: R package `parameters`

Allows the creation of p -value functions directly from models using the function `p_function`. Here is an example from a linear regression model:

► Code



Take-Home-Messages

- The practice of dichotomizing results into “significant” and “not significant” is not informative ([Gelman & Stern, 2006](#)).
- Focus on effect sizes and confidence/compatibility intervals.
- *P*-value functions:
 - are more informative than commonly used summaries of study results.
 - require minimal training because they show familiar measures (*p*-values and confidence intervals).

Take-Home-Messages

- Create p -value functions to summarize the available evidence and improve statistical reasoning:
 - Values with high and low compatibility with the data.
 - How informative was the study?
 - To compare evidence from different studies
 - Useful for meta-analyses (see [Held et al., 2025](#); and [Rücker & Schwarzer, 2021](#)).
- Never say that “we found no effect/association” or “there was no difference” when $p > 0.05$ ([Greenland et al., 2016](#)).

Recommended reading

- Bender, R., Berg, G., & Zeeb, H. (2005). Tutorial: Using Confidence Curves in Medical Research. *Biometrical Journal*, 47(2), 237–247. <https://doi.org/10.1002/bimj.200410104>
- Gelman, A., & Stern, H. (2006). The Difference Between “Significant” and “Not Significant” is not Itself Statistically Significant. *The American Statistician*, 60(4), 328–331. <https://doi.org/10.1198/000313006X152649>
- Greenland, S. (2012). Nonsignificance Plus High Power Does Not Imply Support for the Null Over the Alternative. *Annals of Epidemiology*, 22(5), 364–368. <https://doi.org/10.1016/j.annepidem.2012.02.007>
- Greenland, S. (2019). Valid P-Values Behave Exactly as They Should: Some Misleading Criticisms of P-Values and Their Resolution With S-Values. *The American Statistician*, 73(sup1), 106–114. <https://doi.org/10.1080/00031305.2018.1529625>
- Greenland, S., Senn, S. J., Rothman, K. J., Carlin, J. B., Poole, C., Goodman, S. N., & Altman, D. G. (2016). Statistical tests, P values, confidence intervals, and power: A guide to misinterpretations. *European Journal of Epidemiology*, 31(4), 337–350. <https://doi.org/10.1007/s10654-016-0149-3>
- Held, L., Hofmann, F., & Pawel, S. (2025). A comparison of combined p-value functions for meta-analysis. *Research Synthesis Methods*, 16(5), 758–785. <https://doi.org/10.1017/rsm.2025.26>
- Infanger, D., & Schmidt-Trucksäss, A. (2019). P value functions: An underused method to present research results and to promote quantitative reasoning. *Statistics in Medicine*, 38(21), 4189–4197. <https://doi.org/10.1002/sim.8293>
- Poole, C. (1987a). Beyond the confidence interval. *American Journal of Public Health*, 77(2), 195–199. <https://doi.org/10.2105/AJPH.77.2.195>
- Poole, C. (1987b). Confidence intervals exclude nothing. *American Journal of Public Health*, 77(4), 492–493. <https://doi.org/10.2105/AJPH.77.4.492>
- Rafi, Z., & Greenland, S. (2020). Semantic and cognitive tools to aid statistical science: Replace confidence and significance by compatibility and surprise. *BMC Medical Research Methodology*, 20(1), 244. <https://doi.org/10.1186/s12874-020-01105-9>
- Rosenthal, R., & Rubin, D. B. (1994). The Counternull Value of an Effect Size: A New Statistic. *Psychological Science*, 5(6), 329–334. <https://doi.org/10.1111/j.1467-9280.1994.tb00281.x>
- Rücker, G., & Schwarzer, G. (2021). Beyond the forest plot: The drapery plot. *Research Synthesis Methods*, 12(1), 13–19. <https://doi.org/10.1002/jrsm.1410>